



# Optimization of Inlet–Exhaust Fan Configurations for Vertical Hydroponics Using Computational Fluid Dynamics and Grey Relational Analysis

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**Abstract:** Effective ventilation is essential for maintaining suitable airflow and thermal conditions in controlled-environment vertical hydroponic systems. However, ventilation performance is strongly influenced by the number and arrangement of inlet and exhaust fans, and maximizing airflow velocity alone may not provide an optimal operating environment. In this study, an integrated computational fluid dynamics (CFD) and multi-criteria decision-making framework was developed to optimize inlet–exhaust fan configurations in a vertical hydroponic system accommodating 18 leafy vegetable plants. Ten ventilation configurations were evaluated using three-dimensional CFD simulations, with pressure, airflow velocity, temperature, and computational time considered as performance criteria. Across the simulated configurations, computational times ranged from approximately 160 to 190 s, stabilized temperatures remained near 293 K, airflow velocities ranged from 3.56 to 11.26 m/s, and pressures ranged from 101,325 to 101,408 Pa. The simulated configurations were subsequently evaluated using grey relational analysis, with criterion weights determined using the analytic hierarchy process (AHP). Sensitivity analysis was additionally performed. Under the selected weighting scheme, Configuration 1, comprising one inlet fan and one exhaust fan, achieved the highest grey relational grade and was therefore identified as the preferred ventilation configuration. The results indicate that ventilation effectiveness is governed by the combined performance of airflow and thermal characteristics rather than by maximum airflow velocity alone. Accordingly, increasing the number of fans does not necessarily improve overall ventilation performance. The proposed CFD–AHP–grey relational analysis framework provides a systematic basis for selecting ventilation configurations in vertical hydroponic systems and may support the development of more thermally stable and operationally efficient controlled-environment cultivation systems.

**Keywords:** Vertical hydroponics; Computational fluid dynamics; Ventilation optimization; Grey relational analysis; Analytic hierarchy process; Fan configuration

## 1 Introduction

With the growing population, urbanization, climate variability, and scarcity of land and water, there is a renewed interest in controlled-environment agriculture to achieve better environmental control and space utilization of the crop [1, 2]. Vertical farming and hydroponics are especially appealing due to the ability to produce crops in a limited area, decreasing the need for arable land, and the ability to manage the delivery of water and nutrients [3]. The benefits of vertical systems of cultivation, however, are highly dependent on the ability to create a desirable and consistent crop microclimate [4]. Enclosed growing areas include a continuous interaction among temperature, relative humidity, airflow, concentration of carbon dioxide, and how these factors affect crop transpiration, gas exchange, disease development and crop performance. Environmental regulation is an important technical performance and sustainability aspect of indoor cultivation systems, as highlighted in recent research on controlled-environment agriculture and vertical farming [5–7]. Additionally, control of airflow is critical due to the high vertical stacking and compact geometry of indoor growing structures that can create significant spatial temperature and airflow gradients.

The vertical hydroponic system is especially effective because it includes multiple cultivation levels in a fairly

small enclosure, which makes ventilation more critical [7]. In contrast to open agricultural production systems, the geometry of the enclosure, as well as obstructions inside the system, the position of the fan, the inlet/exhaust layout, and the combined effect of forced convection and thermal gradients play a significant role in limiting airflow within these systems. If the ventilation is not designed properly, stagnant areas, thermal stratification, excessive local velocities and short circuiting between the inlet and exhaust openings can be created. On the other hand, it is not always true that simply increasing the air flow capacity can certainly enhance environmental uniformity. Research in controlled atmospheric conditions has shown that the internal ventilation opening layout and airflow paths can significantly affect internal velocity and temperature profiles and internal structures and crop layouts can create microclimates that vary in space [8, 9]. The results show that, in addition to airflow level, air distribution and direction are important parameters to take into account in designing a ventilation system in compact cultivation systems.

These spatially distributed airflow and thermal phenomena can be effectively studied by the use of computational fluid dynamics (CFD) which is a numerical tool. With CFD, the pressure, velocity and temperature distributions can be calculated in the entire computational domain and give much more information than measurements from a limited number of monitoring points. The use of CFD in greenhouse and controlled-environment agriculture has grown over the past few years to assess the ventilation, circulation of air, temperature distribution and microclimate behavior [8–11]. The vertical arrangement of growing levels, enclosure surfaces and air ventilation can generate complex flow paths within the building, which are difficult to describe by only measuring the global airflow. CFD has also recently been shown as a useful tool to gain insights into the environment of vertical farms in the interior. Research on thermo-fluid-dynamic behavior in an indoor vertical farming system has shown that the local environment is dependent on the air flow distribution and the positioning of the fans [12–15]. In the same way, recent research of vertical farming has also highlighted the need for managing airflow and creating microclimates as technical challenges of vertically integrated production systems [4, 16, 17]. These studies, however, have also shown the need to design the fans for the different specific conditions of a given system, depending on geometry, crop design, airflow needs, and fan configuration.

An important design factor with enclosed hydroponic systems is the inlet/exhaust fan setup. The more inlet fans that are added on, the more momentum that can be imparted to the incoming air and the more the internal velocity field can be altered, but too much airflow at the inlet can lead to higher velocity gradients and higher pressure build-up in the region. Expanding exhaust capacity can allow air to be removed and pressure to be relieved, but an improper inlet/exhaust ratio can create sub-optimal flow paths or short circuiting between the inlet and exhaust areas. The location of the fans is also significant as it changes the preferred airflow path through the enclosure. Therefore, the number of inlet fans, the number of exhaust fans and their arrangement should be taken as a combined problem of ventilation design instead of independent variables. The study of ventilation systems has also shown that the ventilation performance and the resulting internal flow field are affected by the operating conditions of the fans and the air distribution in the building [18, 19].

Another difficulty is the fact that there are several response parameters for ventilation performance and none of them can represent it adequately [20–22]. For instance, to make the velocity of the air as large as possible, the airflow velocity may be high, but this could also cause greater pressures between the airflow or lead to the generation of undesirable airflow patterns. A similar situation applies to the scenario where there is a favorable temperature configuration that needs extra fans, and hence more system complexity. The present problem involves several competing performance criteria such as pressure, velocity of airflow, temperature and computational time. The most optimal configuration for one response may not be the most optimal configuration for the totality of responses.

Grey relational analysis can be used to solve multi-response decision problems due to the capability of transforming different scaled performance characteristics to normalized ones and a grey relational grade to rank the competing alternatives [23–25]. Weighting can be included in the grey relational analysis procedure when there are varying levels of importance to the different performance criteria. The analytic hierarchy process (AHP) is a structured method used to determine the relative importance of the various criteria involved in the decision-making process by comparing them in pairs and calculating normalized priority weights [26]. The use of the AHP and grey relational analysis therefore offers a systematic process for incorporating preferences of the decision makers as well as the quantitative differences that come from CFD simulations. Both the weights of the criteria and the grey relational analysis distinguishing coefficient can, however, affect the final order. Therefore, sensitivity analysis is needed to check the robustness of the chosen configuration.

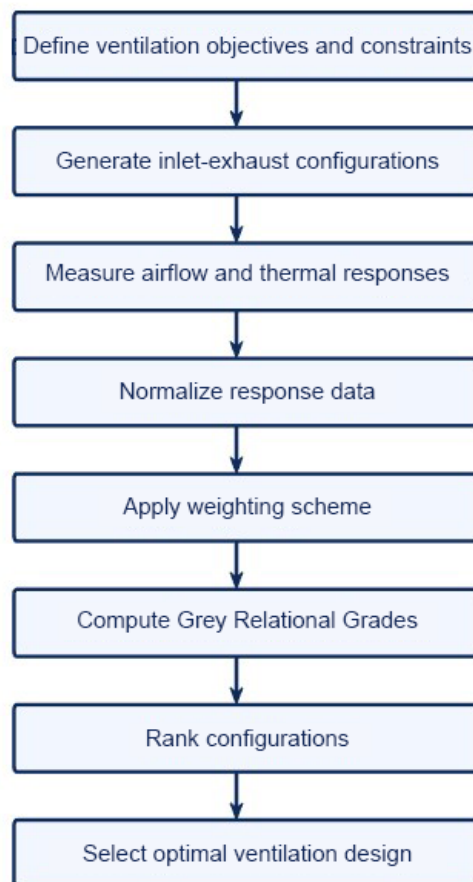
Although inlet–exhaust fan configuration optimization is an emerging and promising research direction for greenhouse, controlled-environment, and vertical-farming applications, relatively little attention has been paid to optimizing inlet–exhaust fan configurations for a compact vertical hydroponic enclosure considering the three parameters of number of inlet fans, number of exhaust fans, and their spatial configuration from a multi-criteria perspective. Previous research on CFD has yielded important results on the distribution of the air flow and thermal field, but the selection of the configurations is generally made based on some individual criteria or simply the

qualitative interpretation of the flow field [27–29]. This leaves a research void in the ability to reduce the detailed CFD outputs to an appropriate quantitative decision-making framework that can be used to rank a set of alternative ventilation configurations that meet conflicting performance criteria.

To fill this void, the present study is presented to investigate ten inlet–exhaust fan configurations in a vertical hydroponic system that can be used for 18 leafy vegetables by using the CFD–AHP–grey relational analysis approach. The three-dimensional CFD simulations are carried out by the software SOLIDWORKS 2021 Flow Simulation, which allows for the definition of the pressure, airflow velocity, computational time and temperature for the configurations that are investigated. They vary in the inlet and exhaust fan locations and fan numbers, and thus allow a systematic investigation of the effects of both the ventilation capacity and the arrangement of the airflow path. The AHP is used to determine the relative weights of the chosen performance criteria and the grey relational analysis combines the weighted CFD responses to produce a combined ranking of the ten configurations. The sensitivity analysis is then performed to explore the impact of inlet and exhaust fan numbers and to gauge the level of robustness of the grey relational analysis ranking given the distinguishing coefficient. The goal is then to find a ventilation configuration that allows for a good relationship between the pressure behavior, the velocity of air flow, the thermal performance and the computational efficiency of the fan system, while avoiding unnecessarily higher fan-system complexity.

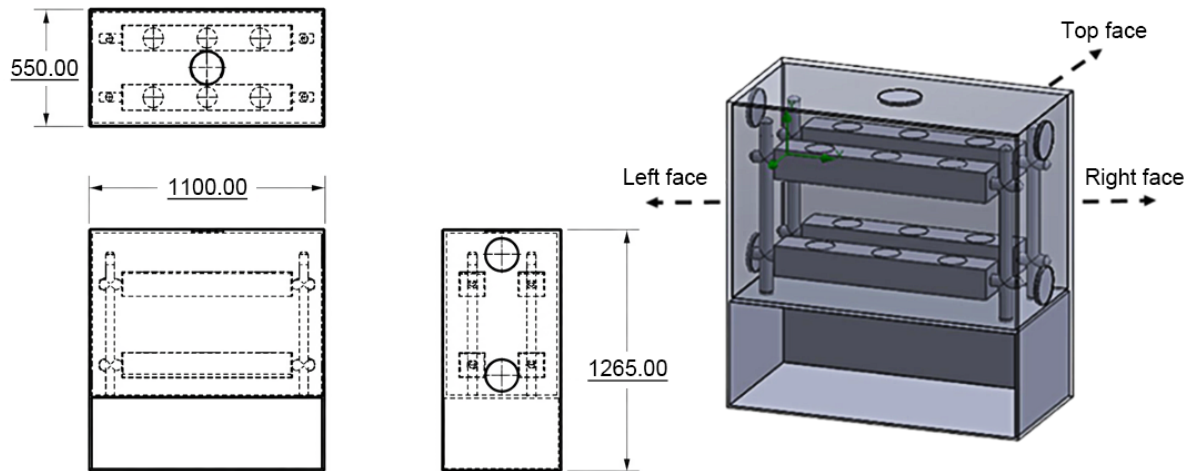
## 2 Methodology

In the present study, a multi-criteria decision framework based on the CFD technique is used to assess and compare different ventilation configurations for a compact vertical hydroponic system with 18 leafy vegetables. The methodology is a combination of three-dimensional CFD simulations, the AHP, and grey relational analysis, aiming to analyze the conflicting influence of inlet and exhaust fan numbers and location. The overall procedure is summarized in the overall block representation of the proposed methodology in Figure 1, which also shows the sequence of the computational and decision-making steps involved, starting with defining the physical system and ending with final ranking with regard to the results obtained.



**Figure 1.** Process flow diagram of the proposed optimization methodology

The geometry was developed physically for the vertical hydroponic enclosure and the ventilation setup. The system was designed for 18 crops of leafy vegetables, using a vertically arranged hydroponic plant cultivation concept. The exterior dimensions and physical configuration of the developed system are shown in Figure 2, which also shows schematic views of the top, right and left surfaces for positioning the ventilation ports and inlet/exhaust fans. All simulations were conducted using the same enclosure geometry to be able to assess the effect of ventilation configuration without altering the geometry. The variables of the ventilation design were the number of inlet fans, the number of exhaust fans and their relative locations. One to four inlet fans were used, while one or two exhaust fans were used. Two different exhaust configurations were explored to account for different paths of air flow through the enclosure: side-mounted and top-mounted. Ten different ventilation configurations were produced, as summarized in Table 1.



**Figure 2.** External dimensions and physical layout of the vertical hydroponic system designed for 18 leafy vegetables

**Table 1.** Inlet and exhaust fan positions for the investigated ventilation configurations

| Configuration | Inlet Fan |       |     | Exhaust |     |
|---------------|-----------|-------|-----|---------|-----|
|               | Left      | Right | Top | Right   | Top |
| 1             | 1         | 0     | 0   | 1       | 0   |
| 2             | 1         | 0     | 0   | 2       | 0   |
| 3             | 2         | 0     | 0   | 1       | 0   |
| 4             | 2         | 0     | 0   | 2       | 0   |
| 5             | 2         | 0     | 1   | 1       | 0   |
| 6             | 2         | 0     | 1   | 2       | 0   |
| 7             | 2         | 0     | 2   | 1       | 0   |
| 8             | 2         | 0     | 2   | 2       | 0   |
| 9             | 2         | 2     | 0   | 0       | 1   |
| 10            | 2         | 2     | 0   | 0       | 2   |

Note: “0” indicates that no fan is installed at the specified location, whereas “1” and “2” indicate one and two fans, respectively.

**Table 2.** Specifications of the fan used in CFD simulations

| Parameter     | Specification                                              |
|---------------|------------------------------------------------------------|
| Manufacturer  | Sanyo Denki America Inc.                                   |
| Model/series  | San Ace 127                                                |
| Airflow       | 170 cubic feet per minute (CFM) (4.76 m <sup>3</sup> /min) |
| Rated voltage | 12 V DC                                                    |

Note: CFD = computational fluid dynamics.

The CFD simulations were carried out using SOLIDWORKS 2021 Flow Simulation. Air was considered as the working fluid, while the internal volume of the hydroponic enclosure was defined as the computational fluid domain. The governing equations of mass, momentum and energy were numerically solved to get the airflow, pressure and thermal fields in the enclosure. All configurations used the same fan to provide a fair comparison. The detailed fan specifications in the simulations are summarized in Table 2. The selected tube-axial fan has a rated voltage of 12 V DC, airflow capacity of 170 cubic feet per minute (CFM) ( $4.76 \text{ m}^3/\text{min}$ ), static pressure of 155 Pa, rotational speed of 3850 rpm and rated power of 16.8 W. The operating characteristics of the fan were therefore not considered for optimization, and the number and spatial configuration of the identical fans were considered as the ventilation design variables.

Each of the ten configurations was developed by systematically varying the number and placement of inlet and exhaust fans while maintaining the same fan characteristics and enclosure geometry. Configurations 1–4 employ inlet fans exclusively on the left surface, with one or two exhaust fans positioned on the right surface. Configurations 5 and 6 retain two inlet fans on the left surface and introduce one additional inlet fan on the top surface, resulting in three inlet fans, while the number of right-surface exhaust fans is varied between one and two, respectively. Configurations 7 and 8 further increase the number of inlet fans by using two fans on the left surface and two on the top surface, giving a total of four inlet fans, with one and two right-surface exhaust fans, respectively. Configurations 9 and 10 use two inlet fans on both the left and right surfaces, giving four inlet fans in total, while one and two exhaust fans, respectively, are positioned on the top surface. Thus, the ten configurations provide systematic variations in both fan number and location, enabling the influence of inlet–exhaust arrangement on the CFD responses to be evaluated consistently (Table 1).

The same conditions were applied at the boundaries of the CFD model in all 10 configurations. The inner box was considered as a computational domain and specified characteristics of the fan were given at the respective inlet and exhaust. All initial and environmental conditions were the same for all configurations. The model was numerically solved with the same numerical settings as given in SOLIDWORKS 2021 Flow Simulation and the fields were evaluated with the same result definition set for all the configurations. The final setting for the simulation should be clearly reported with the CFD assumptions such as the working fluid and its thermos-physical properties, wall thermal condition, initial temperature, fan boundary condition, gravity condition, turbulence treatment, heat-transfer formulation, computational mesh and convergence criteria. Modelling simplifications were made for any components of the CFD domain which were not explicitly represented such as detailed plant-canopy resistance, or leakage through unintended openings. The assumptions were made so that the effect of the configuration of the ventilation could be isolated and a consistent comparison between the ten designs could be made.

The static pressure, the velocity of airflow, the temperature and the time of central processing unit computation were taken as response variables in the CFD simulation. The pressure condition caused by the balance between supplied and exhausted air was thought to be static pressure. The velocity of airflow was used as a parameter to describe the intensity of the airflow in the chamber and its ability to carry heat and gases within the growing area. The decision of temperature was made to check the thermal condition produced by each ventilation system. To ensure the temperature values are comparable, the same SOLIDWORKS result definition was used for all the configurations. The definition of the reported temperature (volume-average, surface-average, maximum temperature, or other specified statistical measure) should be clearly defined based on the actual result quantity that is selected in SOLIDWORKS. Initial conditions for the simulations were an absolute pressure of 101,325 Pa and a temperature of 293.2 K, plus an initial velocity of zero in the X, Y and Z directions. Air was taken as the working fluid. The simulations used the external-flow formulation, and the external walls were assumed to be adiabatic and there was no prescribed heat source. The boundary and initial conditions were the same for all ten configurations and only the number and location of the inlet and exhaust fans were changed.

This important correction has been made to the fourth response variable based on the comments from reviewers. According to the original manuscript, the 160–190 s values correspond to the computational time of the central processing unit and not necessarily to the time the airflow or the temperature field takes to stabilize. As a consequence, the definition of “central processing unit computational time” in the CFD response definition, the CFD formulation (grey relational analysis), and the tables and interpretation are omitted. The computational time on the central processing unit is the time needed by the numerical solver to solve the CFD problem, and is thus treated as a computational-efficiency criterion. It is not to be confused with transient thermal or airflow central processing unit computational time. A true computation at a physical central processing unit level would require an explicit physical time convergence criterion, but would not be possible in the present computation comparison.

These CFD results were then compiled in a decision matrix with the ten configurations. Grey relational analysis was carried out on the normalized responses as the responses are in different physical units and have different preferred directions. Static pressure and temperature and central processing unit computational time were all considered a “smaller the better” criterion while the airflow velocity was considered a “larger the better” criterion. For a smaller-the-better criterion, the normalized response was computed as follows:

$$X_i^* = \frac{\max(X_i(k)) - X_i(k)}{\max(X_i(k)) - \min(X_i(k))} \quad (1)$$

$$X_i^* = \frac{X_i(k) - \min(X_i(k))}{\max(X_i(k)) - \min(X_i(k))} \quad (2)$$

where,  $X_i^*(k)$  is the normalized value of the  $i$ -th configuration for the  $k$ -th criterion,  $X_i(k)$  is the original value of the  $i$ -th configuration for the  $k$ -th criterion, and  $\max(X_i(k))$  and  $\min(X_i(k))$  are the maximum and minimum values of the  $k$ -th criterion across all configurations. The normalization converts units of the different types of responses into a dimensionless scale ranging from 0 to 1, permitting the four CFD responses to be assessed in the same decision context.

The four performance criteria were then mapped to the AHP and the relative importance of the criteria was determined prior to their integration into the grey relational analysis. The criterion hierarchy was the overall goal of selecting the most favored ventilation configuration, followed by the four CFD response criteria, namely temperature, central processing unit computational time, airflow velocity, and static pressure. The nine-point preference scale was used in the conventional approach for creating pair-wise comparisons. The judgments were determined based on the order of the functional priorities of the ventilation system and not as a result of a separate expert survey. Temperature was the most important parameter as thermal regulation is directly related to the desired cultivation environment. Airflow velocity and static pressure were ranked second and third, respectively, in terms of their importance for the computational time of the central processing unit. Table 3 shows the AHP pairwise comparison matrix used to determine the weights.

**Table 3.** AHP pairwise comparison matrix for the CFD performance criteria

| Criteria               | Temperature | Computational Time | Airflow Velocity | Static Pressure |
|------------------------|-------------|--------------------|------------------|-----------------|
| Temperature            | 1           | 1                  | 2                | 2               |
| CPU computational time | 1           | 1                  | 1                | 2               |
| Airflow velocity       | 1/2         | 1                  | 1                | 2               |
| Static pressure        | 1/2         | 1/2                | 1/2              | 1               |

Note: AHP = analytic hierarchy process; CFD = computational fluid dynamics; CPU = central processing unit.

The AHP pairwise comparison matrix was established by assigning relative importance to the four CFD performance criteria according to their intended role in evaluating the ventilation configuration. Temperature was assigned the highest relative importance because thermal conditions are directly relevant to the controlled hydroponic environment, followed by computational time, airflow velocity, and static pressure. The principal-eigenvector method was used to derive the normalized priority weights. The resulting weights were 0.3397 for temperature, 0.2808 for computational time, 0.2390 for airflow velocity, and 0.1404 for static pressure. The consistency index was 0.0202 and the consistency ratio was 0.0225, which is below the 0.10 criterion, confirming acceptable consistency of the pairwise judgments.

After the AHP weighting, the grey relation coefficient was determined for every configuration and every response. The coefficient was calculated as follows:

$$\xi_i(k) = \frac{\Delta_{\min} + \zeta \Delta_{\max}}{\Delta_i(k) + \zeta \Delta_{\max}} \quad (3)$$

where,  $\xi_i(k)$  represents the absolute deviation between the ideal reference sequence and the normalized response of configuration  $i$ , and  $\Delta_{\min}$  and  $\Delta_{\max}$  are the minimum and maximum values of the deviation sequence, respectively. The parameter  $\zeta$  is the distinguishing coefficient. The value of 0.5 was originally used for the base grey relational analysis calculation. However, since the value of the grey relational analysis can affect the relational coefficients and ranking, the effects of this value were subsequently investigated using the sensitivity analysis.

Then the weighted grey relational grade was calculated as follows:

$$\gamma_i = \sum_{j=1}^m w_k \xi_i(k) \quad (4)$$

where, the weight of the criterion  $j$ ,  $w_k$ , is derived based on the AHP and the grey relational coefficient,  $\xi_i$  is the corresponding coefficient, and  $m$  is the number of the response criteria. The resulting grey relational grade is a

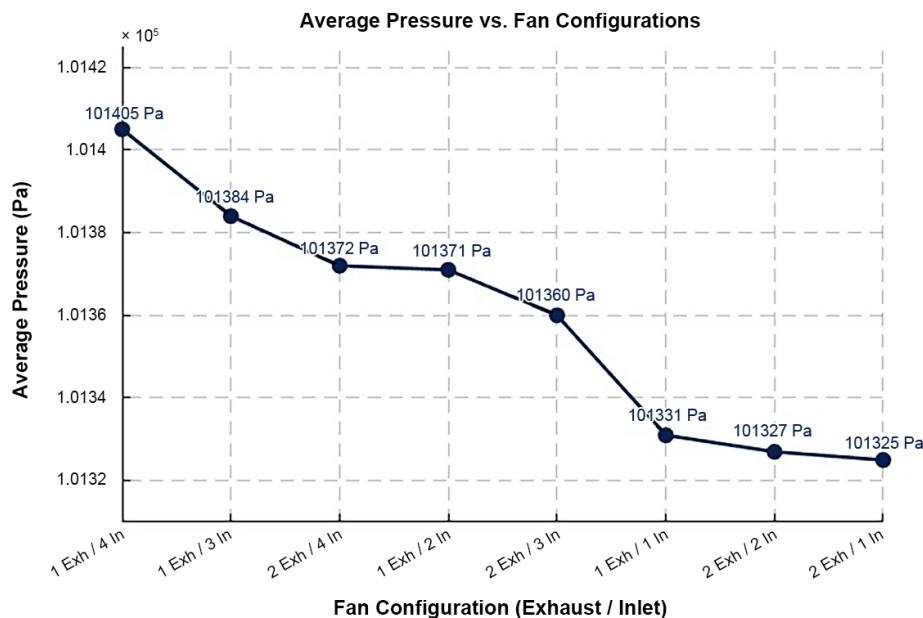
single integrated performance indicator for each ventilation configuration. A positive grey relational grade means that there is a generally closer match with the desired reference condition given the relative importance of the four performance criteria.

The number of fans was considered as a design variable and not as a separate criterion of the grey relational analysis. Therefore, the number of fans is not considered as an extra weighted response in Eq. (4). Instead, the effect of inlet and exhaust fan numbers was assessed using the configuration matrix and sensitivity analysis. This separation ensures that the responses of the CFD are not repeated to account for the complexity of the system, but that the effect of the amount and location of fans on the CFD can be investigated separately. To explore the impact of inlet and exhaust fan numbers on the CFD responses, sensitivity analysis was carried out. The individual responses were plotted as main-effect plots to illustrate the influence of changes in ventilation capacity on the individual response. The main-effect analysis offers an alternative explanation of the CFD results to determine if the inlet-fan number, the exhaust-fan number, or a combination of both due to the airflow pathway had a larger impact on the responses. Sensitivity results were then employed to aid the interpretation of the AHP weighting scheme instead of supplanting the performance measures derived from the CFD.

A second sensitivity analysis was carried out to evaluate the robustness of the grey relational analysis ranking with respect to the distinguishing coefficient. The grey relational analysis calculation was repeated for a number of different values of the distinguishing coefficient, instead of using only  $\zeta = 0.5$ . For every value of  $\zeta$ , a list of the grey relational grade and ranking of all ten configurations was recorded. If the ranking of the preferred configuration did not change over the range of investigation, it was deemed robust. This analysis directly tackles the potential dependence of the final decision on the conventional selection of the distinguishing coefficient.

### 3 Results and Discussion

The effects of inlet-fan number, exhaust-fan number, and fan position on air flow and thermal properties of the vertical hydroponic enclosure were examined using the CFD simulations. The effect of ventilation configuration was examined without altering the geometric configuration of the enclosure or the characteristics of the fan, by testing ten different configurations. The air flow configurations investigated consist of one or two exhaust fans, and one to four inlet fans, with the exhaust fans either placed on the right surface or on the top surface of the device as described in Table 1. This presentation is based on the configuration of the inlet and exhaust fans, which is significant because the simulated fan response depends not only on the number of inlet fans but also on the relative inlet–exhaust configuration and, consequently, the inlet flow path traversed.

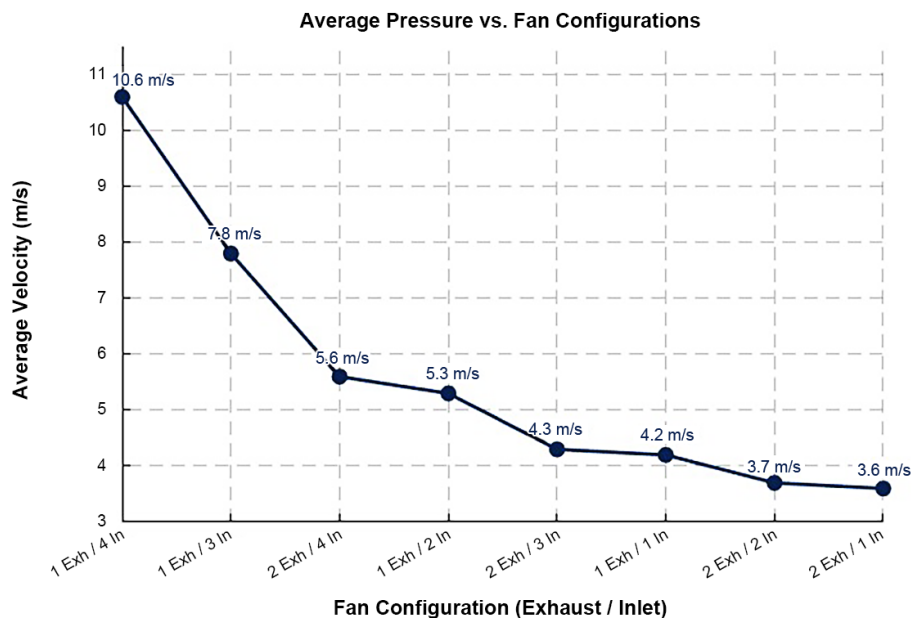


**Figure 3.** Pressure response with respect to exhaust and inlet fans

The static-pressure results clearly indicate that there is a strong dependence on inlet/exhaust ratio. The pressure values on the ten configurations (Figure 3) are between 101,325 Pa and 101,408 Pa. More inlet fans typically can result in higher pressure, especially when the number of exhaust fans is kept low. Both Configurations 7 and 9 show four inlet fans with a single exhaust fan which yield significantly higher pressures, 101,401.20 and 101,408.12 Pa, respectively. As inlet capacity is increased, internal pressurization tends to increase when there is no increase in the

exhaust capacity, as seen in the comparison. On the other hand, increasing the exhaust capacity tends to reduce the internal pressure. For instance, with Configuration 8 (four inlet fans and two exhaust fans), the pressure is 101,371.10 Pa, whereas the configuration with four inlet fans and one exhaust fan produces a higher pressure of 101,408.12 Pa. The results show that the pressure is influenced not only by the inlet capacity but also by the balance between the airflow entering through the inlet fans and that discharged through the exhaust fans. Therefore, the more fans installed, the better it cannot be taken for granted when there is not enough exhaust capacity.

The airflow-velocity results add to the evidence of the impact of the ventilation configuration. The simulated velocity can be seen to increase significantly with the increase of inlet-fan capacity, as shown in Figure 4, up to 11.26 m/s in Configuration 9 with four inlet fans and one exhaust fan. Four inlet fans and one exhaust fan are again used for Configuration 7 and the velocity of 10.009 m/s is a relatively high value. For the other configurations, Configuration 1 has a maximum velocity of 3.556 m/s and Configuration 4 (with two inlet and two exhaust fans) has a velocity of 3.694 m/s. The comparison shows that the inlet fans increase the airflow velocity; however, a greater increase in velocity does not necessarily correspond to improved ventilation effectiveness. If the exhaust system is not designed to allow air to be removed efficiently, then high velocity can be present with non-uniform flow distribution. The addition of a second exhaust fan at similar inlet capacity further illustrates the influence of exhaust capacity on the flow field since velocity decreases with the addition of the second fan. The velocity results confirm the interpretation that ventilation quality is more related to the interaction of inlet supply and exhaust removal than to simply increasing the inlet airflow.



**Figure 4.** Velocity response with respect to exhaust and inlet fans

The temperature response is a complementary view of the ventilation performance. As can be seen in Table 4, the temperatures reported are very close to each other for all the configurations except Configuration 1, meaning that the differences among Configurations 2–10 are relatively small. Each configuration reports a different value of the temperature: Configuration 1 reports a temperature of 303.16 K, whereas Configurations 2–10 report values of approximately 293.16 K. This difference is sufficiently large to assume a ventilation-related trend unless taking into account the temperature metric extracted from SOLIDWORKS Flow Simulation. Thus, the revised analysis does not claim to explain the difference because the result definition and CFD set-up have not been verified. Nevertheless, the reported value is retained in Table 4 and Figure 5. Assuming that the temperature extraction method of the various configurations is identical, the difference can be compared with the respective airflow field. In Configurations 2–10, the low range of the temperature response suggests that the extra ventilation capacity makes only small differences in the reported temperature responses. It can be observed that, beyond a certain number of fans, further increases in the number of fans do not necessarily result in proportional improvements in temperature.

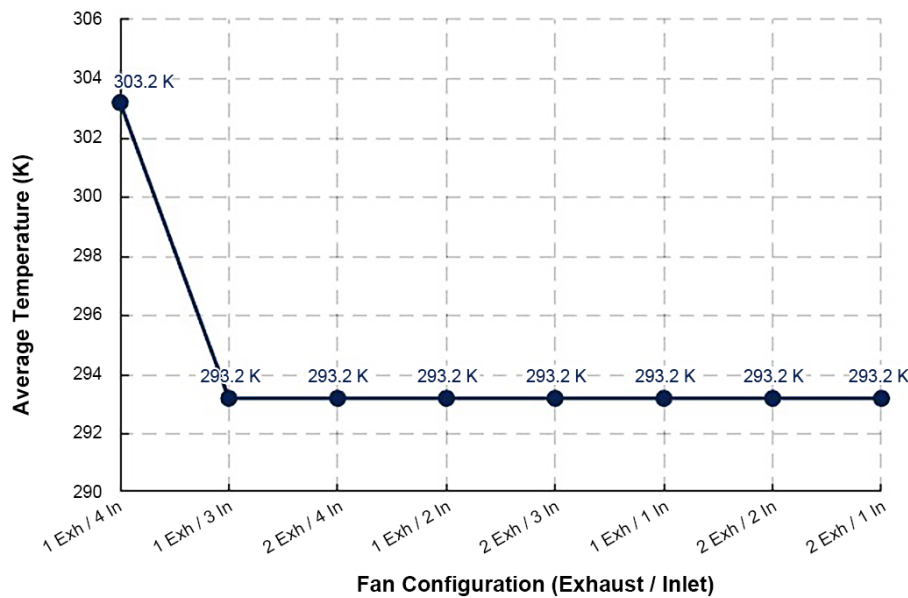
The values reported in Table 4, ranging from 160 to 190 s, represent the central processing unit computational time required by SOLIDWORKS 2021 Flow Simulation, rather than the physical central processing unit computational time of airflow or temperature within the hydroponic chamber. Configuration 1 required 160 s of computational time, while Configurations 9 and 10 required 190 s. The variation therefore reflects differences in the numerical solution process associated with the respective CFD models and should not be interpreted as evidence that one configuration

physically reaches thermal equilibrium faster than another. Figure 6 is consequently interpreted as a comparison of computational time rather than transient stabilization behavior. This distinction is important because computational runtime and physical response time represent fundamentally different quantities. A physical stabilization-time analysis would require transient CFD simulations with an explicitly defined physical-time criterion, which was not the basis of the present steady-state comparative analysis.

**Table 4.** CFD-derived performance indicators, including average velocity, pressure, temperature, and central processing unit computational time

| Configuration | Number of Inlet Fans | Number of Exhaust Fans | Pressure (Pa) | Velocity (m/s) | Temperature (K) | Computational Time (s) | Details                                                                      |
|---------------|----------------------|------------------------|---------------|----------------|-----------------|------------------------|------------------------------------------------------------------------------|
| 1             | 1                    | 1                      | 101330.73     | 3.556          | 303.16          | 160                    | —                                                                            |
| 2             | 1                    | 2                      | 101325        | 4.192          | 293.2           | 170                    | —                                                                            |
| 3             | 2                    | 1                      | 101370.87     | 5.305          | 293.23          | 170                    | —                                                                            |
| 4             | 2                    | 2                      | 101326.72     | 3.694          | 293.21          | 180                    | Two exhaust fans on the right                                                |
| 5             | 3                    | 1                      | 101384.2      | 7.798          | 293.24          | 175                    | Two inlet fans on the left and one on the top; one exhaust fan on the right  |
| 6             | 3                    | 2                      | 101359.72     | 4.25           | 293.23          | 180                    | Two inlet fans on the left and one on the top; two exhaust fans on the right |
| 7             | 4                    | 1                      | 101401.2      | 10.009         | 293.24          | 175                    | Two inlet fans on the left and two on the top; one exhaust fan on the right  |
| 8             | 4                    | 2                      | 101371.1      | 5.302          | 293.24          | 185                    | Two inlet fans on the left and two on the top; two exhaust fans on the right |
| 9             | 4                    | 1                      | 101408.12     | 11.26          | 293.25          | 190                    | Two inlet fans on the left and two on the right; one exhaust fan on the top  |
| 10            | 4                    | 2                      | 101372.4      | 5.968          | 293.24          | 190                    | Two inlet fans on the left and two on the right; two exhaust fans on the top |

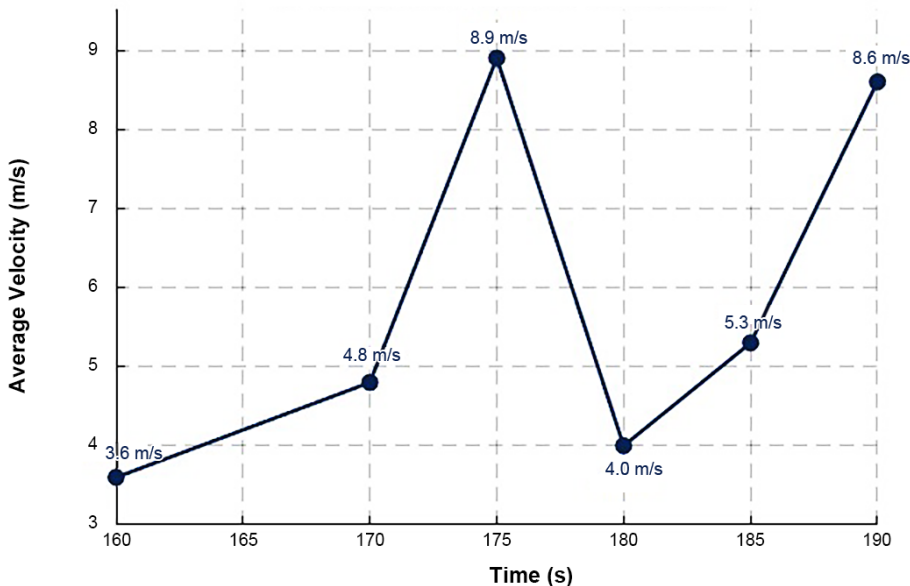
Note: CFD = computational fluid dynamics; “—” denotes no additional details provided.



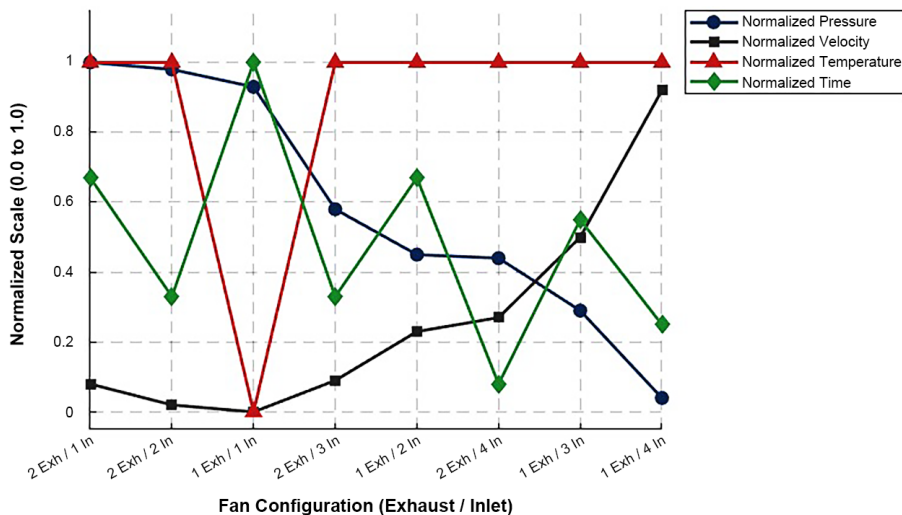
**Figure 5.** Thermal response with respect to exhaust and inlet fans

The normalized response representation in Figure 7 facilitates comparison of the four criteria despite their different units and numerical ranges. Pressure, temperature, and central processing unit computational time were treated as smaller-the-better criteria, whereas airflow velocity was treated as a larger-the-better criterion. The normalized trends demonstrate that increasing the number of inlet fans generally increases airflow velocity and pressure, while the introduction of additional exhaust capacity tends to moderate pressure and airflow velocity. The temperature

response is comparatively less differentiated for Configurations 2–10 because the corresponding simulated values remain close to 293 K. The computational-time trend also does not follow a simple monotonic relationship with fan number, confirming that central processing unit time should not be interpreted as a physical ventilation-response variable. Figure 7 therefore provides a comparative visualization of the normalized CFD responses and illustrates the conflicting nature of the design objectives. A configuration that maximizes velocity may simultaneously increase pressure, while a configuration that minimizes computational time may not provide the desired airflow characteristics.



**Figure 6.** Relationship between computational time (s) and average velocity (m/s) for the ten ventilation configurations



**Figure 7.** Normalized variation of pressure, airflow velocity, stabilized temperature, and central processing unit computational time with different inlet-exhaust fan configurations

The main-effect sensitivity analysis in Figure 8 provides an additional insight into the mechanisms underlying the CFD responses. The inlet fan number greatly affects the pressures and velocities of airflow. An increase in inlet capacity can increase the amount of air entering the enclosure, causing the internal pressure and flow velocity to rise. The velocity response is more pronounced at higher inlet fan levels, as this is where the velocity increases faster. The number of exhaust fans has an inverse effect on pressure as the more fans, the better it will be to remove the air. Thus, the pressure response depends on the ratio of the inlet capacity to the exhaust capacity and is not



interpreted as empirically derived universal importance factors for all hydroponic systems.

The weighted grey relational analysis results offer an overall rank of the ten configurations that take into account the pressure, velocity, temperature and central processing unit computational time. The results in Table 6 indicate that the highest reported grey relational grade is 0.492 for Configuration 1, which consists of one inlet fan and one exhaust fan. The next most popular configuration, at 0.486, is Configuration 10, and Configuration 8 is third with a grey relational grade of 0.482. The result shows that the best configuration is not the one with the highest air flow velocity. Rather, Configuration 1 has the best grade in integrated scores based on the weighting and normalization approach adopted. This lends support to the central concept of this study which is that ventilation selection should be based upon their combination rather than on the magnitude of the airflow.

Prior to calculating the grey relational coefficients, the performance responses were normalized according to their optimization objectives. Airflow velocity was treated as a larger-the-better characteristic, whereas pressure, temperature, and computational time were treated as smaller-the-better characteristics. The normalized values were subsequently used to define the reference sequence and compute the grey relational coefficients and grey relational grades. All calculations were verified to ensure consistency with the selected optimization criteria.

**Table 5.** AHP criterion weights used for the weighted grey relational analysis

| Criterion          | Desired Trend      | Weight | Sensitivity-based Justification                                               |
|--------------------|--------------------|--------|-------------------------------------------------------------------------------|
| Temperature        | Smaller-the-better | 0.33   | Strong thermal sensitivity to ventilation configuration                       |
| Computational time | Smaller-the-better | 0.27   | Represents the computational efficiency of the CFD analysis                   |
| Airflow velocity   | Larger-the-better  | 0.25   | Strong response to inlet-fan number and moderation by exhaust capacity        |
| Static pressure    | Smaller-the-better | 0.15   | Strong dependence on inlet flow with pressure relief through exhaust capacity |

Note: AHP = analytic hierarchy process; CFD = computational fluid dynamics.

**Table 6.** Ranking of configurations based on the grey relational analysis methodology

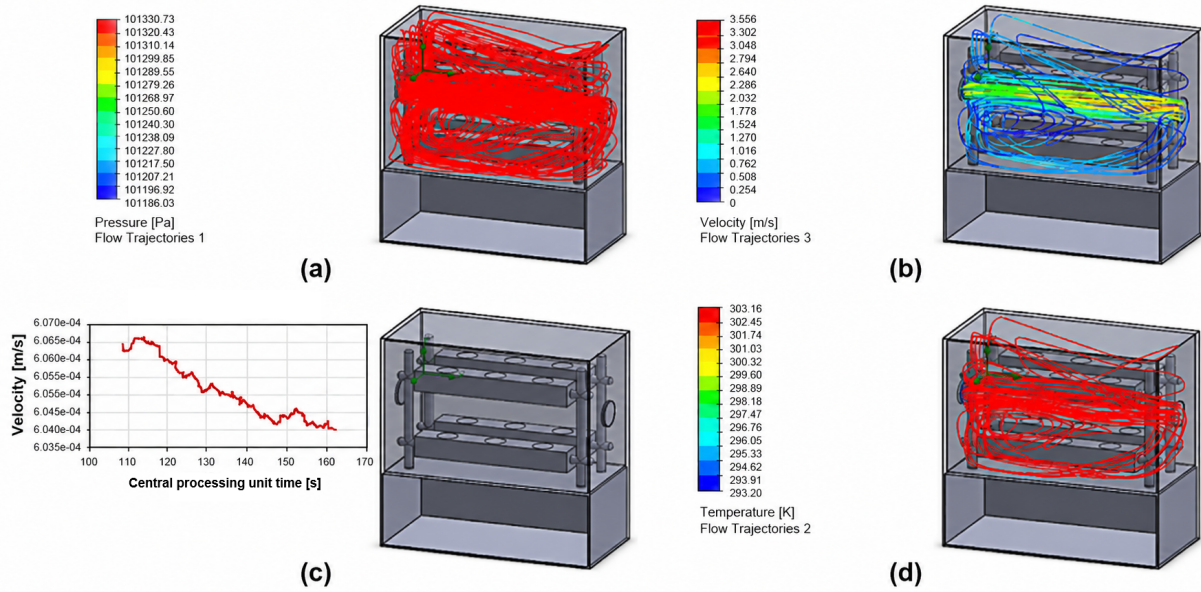
| Configuration    | Number of Inlet Fans | Number of Exhaust Fans | Pressure | Velocity | Temperature | Time | Grey Relational Grade | Rank |
|------------------|----------------------|------------------------|----------|----------|-------------|------|-----------------------|------|
| Criterion weight | —                    | —                      | 0.15     | 0.25     | 0.33        | 0.27 | —                     | —    |
| 1                | 1                    | 1                      | 0.363    | 0.600    | 0.600       | 0.33 | 0.492                 | 1    |
| 2                | 1                    | 2                      | 0.333    | 0.586    | 0.333       | 0.45 | 0.429                 | 10   |
| 3                | 2                    | 1                      | 0.513    | 0.560    | 0.335       | 0.45 | 0.450                 | 6    |
| 4                | 2                    | 2                      | 0.342    | 0.597    | 0.334       | 0.53 | 0.456                 | 5    |
| 5                | 3                    | 1                      | 0.548    | 0.487    | 0.335       | 0.50 | 0.450                 | 7    |
| 6                | 3                    | 2                      | 0.479    | 0.585    | 0.335       | 0.53 | 0.474                 | 4    |
| 7                | 4                    | 1                      | 0.586    | 0.398    | 0.335       | 0.50 | 0.433                 | 9    |
| 8                | 4                    | 2                      | 0.513    | 0.560    | 0.335       | 0.57 | 0.482                 | 3    |
| 9                | 4                    | 1                      | 0.600    | 0.333    | 0.336       | 0.60 | 0.446                 | 8    |
| 10               | 4                    | 2                      | 0.517    | 0.543    | 0.335       | 0.60 | 0.486                 | 2    |

Note: “—” indicates that no criterion weight is applicable to the corresponding column.

The detailed CFD results for Configuration 1 are presented below. Figure 9a shows the spatial distribution of the selected pressure distribution, whereas Figure 9b shows the airflow field of velocity. The computational-time-related CFD result is shown in Figure 9c and is referred to as central processing unit computational time. The temperature distribution is shown in Figure 9d. These field visualizations help to support the scalar values given Table 4 and the integrated ranking in Table 6 with spatial information. They, in particular, enable the selected configuration to be examined not just from the point of view of the numbers of the performance metrics but also the spatial distribution of pressure, velocity and temperature within the enclosure.

Configuration 1 is made up of one inlet fan and one exhaust fan and has the highest reported grey relational analysis grade, although this configuration has the lowest number of fans among the configurations investigated. This is a significant outcome from a system design point of view, as the number of fans not only increases the capacity of the airflow but also alters the pressure field and airflow pathways. For instance, Configurations 7 and 9 have much higher velocities as well as higher inlet capacity, pressure values and lower grey relational analysis ranking

than the lower-velocity Configuration 1. The comparison thus indicates that the optimization of airflow velocity is not equivalent to the optimization of multi-criteria ventilation performance. The grey relational analysis framework, on the other hand, supports the configuration with the best compromise of the chosen responses. The result also highlights how crucial it is to take exhaust capacity into account with inlet capacity. Configurations 1 and 2 have the same number of inlet fans, but Configuration 1 has a lower exhaust capacity than Configuration 2. It reduces the pressure from 101,330.73 to 101,325 Pa, and it generates a temperature of 293.20 K. However, its grey relational analysis grade is lower than that of Configuration 1, according to the adopted weighting. Likewise, Configurations 7 and 8 have four inlet fans and four exhaust fans, respectively, with a difference in exhaust capacity, and the pressure drops to 101371.10 and 5.302 m/s, respectively. The above comparisons show that the exhaust arrangement has a significant effect on the operating state of a given inlet capacity. The ideal configuration is then not based on the fan count, but on the combination of response.



**Figure 9.** Static pressure, airflow velocity, central processing unit computational time and temperature distributions for Configuration 1

These outcomes from the sensitivity analysis are consistent with this interpretation. The inlet-fan number has a significant influence mainly on pressure and velocity of air flow; the exhaust arrangement, on the other hand, determines the pressure drop and influences the airflow pathway. This gives rise to the response interactions which account for the fact that configurations with large airflow magnitudes are not necessarily the ones with the greatest grey relational analysis grade. From a practical design point of view, this means that, when designing a compact vertical hydroponic system, one should look at the total inlet/exhaust loop rather than just adding fans to boost the air circulation. For situations where a lower complexity arrangement offers sufficient pressure control, airflow, thermal behavior and computational performance may be better suited. The result in the CFD field in Figure 9 should therefore be used as supporting evidence for the quantitative ranking, but not as an independent validation of the grey relational analysis result. The results from the numerical simulations illustrate the effect of changing the fan configuration on the internal flow and thermal fields, and the grey relational analysis gives a systematic way of combining the competing responses. The present study is computational and does not contain experimental measurement of airflow velocity or temperature in the physical hydroponic system; therefore, the CFD results are a numerical assessment and cannot be considered to be experimentally validated predictions of airflow velocity or temperature. The selected configuration is then validated in experiments before being generalized to other hydroponics geometries, plant densities, environmental conditions and fan specifications.

The robustness of the grey relational analysis ranking was tested by changing the distinguishing coefficient of the grey relational analysis from 0.1 to 0.9 and keeping the same criterion weights and the same normalization procedure and preference directions. For each  $\zeta$  value, the grey relational grade and ranking of all ten ventilation configurations were re-computed, as shown in Table 7. The fact that Configuration 1 is the most popular option, even when the value of the distinguishing coefficient is set to zero, shows that the chosen ventilation configuration is robust with respect to the distinguishing coefficient and is not only the result of using the conventional value of  $\zeta = 0.5$ .

The results combined suggest that the capacity of the inlet, exhaust capacity and fan position have a combined

effect on the ventilation performance of the vertical hydroponic enclosure investigated. The increased inlet capacity can result in higher airflow velocity; however, when exhaust capacity is lower, it can also result in higher pressure. The effectiveness of the ventilation system cannot be measured by inlet airflow only as there is additional exhaust capacity for pressure relief and changes in the airflow path. The grey relational analysis results show that the combination of one inlet fan and one exhaust fan (Configuration 1) is the most preferred configuration based on the weighting scheme reported. The above result should be viewed as the best outcome for the ten configurations and CFD assumptions explored, and not as a general best outcome for all vertical hydroponic systems. The proposed CFD–AHP–grey relational analysis framework offers a systematic approach to the analysis of ventilation configurations with respect to several conflicting criteria. The main application of the framework is to enable decision makers to analyze the various possible fan configurations, before any physical manufacturing, to account for pressure, velocity of air flow, temperature, and computational needs. The approach can mitigate the need for single-response selection and discover lower-complexity design that meet an acceptable compromise of conflicting performance demands. However, further research is needed that includes experimental CFD validation, humidity and carbon-dioxide transport, plant-canopy resistance, plant-scale microclimate measurements, fan energy consumption, and transient environmental conditions. The extension of the numerical configuration ranking from the level of airflow assessment in the enclosure to experimentally validated, plant-centered ventilation optimization would result from these additions.

**Table 7.** Sensitivity of the grey relational analysis results to the distinguishing coefficient  $\zeta$

| $\zeta$ | C1        | C2         | C3        | C4        | C5        | C6        | C7        | C8        | C9        | C10       |
|---------|-----------|------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 0.1     | 0.173 (1) | 0.137 (10) | 0.145 (7) | 0.151 (5) | 0.144 (8) | 0.158 (4) | 0.137 (9) | 0.164 (3) | 0.150 (6) | 0.167 (2) |
| 0.2     | 0.290 (1) | 0.237 (10) | 0.251 (7) | 0.258 (5) | 0.250 (8) | 0.270 (4) | 0.239 (9) | 0.278 (3) | 0.255 (6) | 0.282 (2) |
| 0.3     | 0.375 (1) | 0.314 (10) | 0.332 (7) | 0.339 (5) | 0.332 (8) | 0.353 (4) | 0.318 (9) | 0.363 (3) | 0.333 (6) | 0.367 (2) |
| 0.4     | 0.440 (1) | 0.377 (10) | 0.396 (7) | 0.402 (5) | 0.397 (6) | 0.419 (4) | 0.381 (9) | 0.428 (3) | 0.396 (8) | 0.433 (2) |
| 0.5     | 0.492 (1) | 0.428 (10) | 0.449 (6) | 0.454 (5) | 0.450 (7) | 0.472 (4) | 0.433 (9) | 0.481 (8) | 0.446 (3) | 0.486 (2) |
| 0.6     | 0.534 (1) | 0.471 (10) | 0.493 (7) | 0.497 (5) | 0.494 (6) | 0.515 (4) | 0.477 (9) | 0.525 (3) | 0.488 (8) | 0.529 (2) |
| 0.7     | 0.570 (1) | 0.508 (10) | 0.530 (7) | 0.533 (5) | 0.531 (6) | 0.552 (4) | 0.514 (9) | 0.561 (3) | 0.524 (8) | 0.565 (2) |
| 0.8     | 0.600 (1) | 0.540 (10) | 0.562 (7) | 0.565 (5) | 0.563 (6) | 0.583 (4) | 0.546 (9) | 0.593 (3) | 0.555 (8) | 0.596 (2) |
| 0.9     | 0.626 (1) | 0.568 (10) | 0.590 (7) | 0.592 (5) | 0.591 (6) | 0.611 (4) | 0.574 (9) | 0.619 (3) | 0.582 (8) | 0.623 (2) |

#### 4 Conclusion and Future Research Directions

The objective of this study is to design a multi-criteria decision framework for the evaluation of the ventilation configuration in a compact vertical hydroponic growing system for 18 leafy vegetables based on CFD simulation. SOLIDWORKS 2021 Flow Simulation was used to study ten configurations with various combinations and locations of inlet and exhaust fans. Performance criteria for comparing the ventilation alternatives were static pressure, airflow velocity, temperature and central processing unit computational time. The results show that more inlet fans do not necessarily ensure higher ventilation performance. The use of extra inlet fans can increase the air velocity to a high degree but at the same time can increase the internal pressure, especially if the exhaust capacity is limited. This comparison between the one and two exhaust fan configurations further demonstrates that exhaust capacity is a factor in controlling the pressure and results in a modifiable airflow pathway. The pressure values calculated for the analyzed configurations ranged from 101,325.00 to 101,408.12 Pa, the air velocities from 3.556 to 11.260 m/s, the temperatures from 293.20 to 303.16 K, and the computational time for the central processing unit from 160 to 190 s. The maximum air flow velocity was found with Configuration 9 (four inlet fans and one exhaust fan) and the lowest air flow velocity was achieved with Configuration 1 (one inlet fan and one exhaust fan) with velocities being 3.556 m/s and 1.278 m/s, respectively. The results show that there is a compromise between the airflow capacity and the control of the enclosure's pressure. The computational-time results are strictly based on central processing unit computational time needed to solve the CFD problem, and should not be interpreted as being based on physical airflow or thermal central processing unit computational time. This distinction is important because the original interpretation is limited by this factor and computational runtime should not be confused with the transient behavior of the hydroponics environment.

For Configuration 1, the temperature response is a different behavior. The maximum temperature of the trajectory, as shown in the flow trajectory results, is 303.16K while for Configurations 2–10, the result is between 293.20K and 293.25K. It is important to note that this difference is not caused by any heat transfer from the enclosure as the initial condition and the treatment of the external walls were set adiabatically, and no heat source was prescribed for the simulations. The value of 303.16 K represents the maximum temperature along the computed flow trajectory rather than the domain-averaged temperature. The computed solution of the flow and energy equation may present a temperature difference from one point to another within the region of the flow due to spatial variation of pressure and

velocity. Configuration 1 also exhibits a greater pressure difference along the displayed trajectories than Configuration 2, which suggests that the pressure field close to the trajectories is significantly different from that in Configuration 2. However, the temperature calculated did not account for crop transpiration, radiation, internal heat generation, or conjugate heat transfer so it should be used as a comparative index for CFD only and not as a complete prediction of the crop canopy thermal environment.

It was found that the multi-criteria analysis results in a different design selection if airflow velocity is the only criterion used to select the design against choosing all performance criteria simultaneously. The relative importance of the four variables, namely temperature, computational time of the central processing unit, airflow velocity, and static pressure was determined by using the AHP, and then the normalized responses were integrated into a single performance grade by using the grey relational analysis. Using the adopted weights of 0.33 for temperature, 0.27 for central processing unit computational time, 0.25 for airflow velocity, and 0.15 for static pressure, the highest reported grey relational grade is 0.492 for Configuration 1, followed by Configuration 10 at 0.486 and Configuration 8 at 0.482. The result shows that the most preferred ventilation arrangement is not the one which understandably results in maximum velocity of airflow but rather the arrangement that results in the most favorable compromise among the selected criteria. The ranking thus justifies the use of a multi-criteria approach for the selection of the ventilation configuration in compact hydroponic systems.

The sensitivity analysis also demonstrated that the number of inlet fans has a significant effect on the airflow velocity and pressure, and the exhaust fan configuration has a significant effect on the modification of the airflow pathway and pressure regulation. This discovery raises an important design implication: Fan quantity cannot be increased without considering the available exhaust pathway. This can achieve a more controlled environment for the airflow inside the machine without the need for unnecessary ventilation hardware by using a balanced inlet/exhaust design. It can be used as a framework to compare the various fan configurations prior to physical installation and can be used as a quantitative measure to select a fan configuration based on a set of performance criteria. The identification of Configuration 1 as the highest-ranked alternative is significant specifically because it uses the lowest number of inlet fans and exhaust fans to obtain the best grey relational analysis grade in the investigated design space, which can reduce the complexity of the design when compared to using three or four inlet fans.

Additional spatial evidence for the interpretation of the chosen configuration is given from the CFD visualizations, where the pressure, velocity and temperature distributions are displayed for the chosen configuration. The present results, however, are not to be taken as a prediction to be validated in experiment but as a comparative numerical assessment. Air flow velocity, pressure and temperature within the physical hydroponic enclosure are not measured in the experiment, so the numerical results cannot be generalized to other enclosure dimensions, plant densities, fan specifications, or environmental conditions. Moreover, the temperature response of the trajectory was different in Configuration 1, reaching a maximum of 303.16 K. The value is a local temperature obtained from the CFD simulation rather than a direct measure of external heat gain or the domain-averaged crop temperature, as no heat sources were prescribed and the enclosure walls were treated as adiabatic.

Another limitation concerns the selection of the AHP weights and the grey relational analysis distinguishing coefficient. The criterion weights are the engineering priorities selected in the present study, and are not intended as universal weights for all vertical hydroponic systems. In the same way, the conventional judging level  $\zeta = 0.5$  may affect the calculated grey relational grade. Hence, it is advisable to conduct sensitivity analysis for the grey relational analysis distinguishing coefficient to assess the robustness of the reported ranking. Humidity, carbon dioxide concentration, plant resistance, evapotranspiration, energy use, and the specific environmental needs of crops are also not explicitly taken into account in the current analysis. When the objective is moved from the enclosure level of airflow optimization to the plant level of growth optimization, these variables may become more significant.

Experimental validation of the CFD prediction should, therefore, be the first step for future research, which should involve physical measurements of air velocity, pressure, and temperature at several sites in the hydroponic enclosure. Experimental measurements must be taken under the same fan configuration that is used in the simulations so that the numerical model can be quantitatively validated, and if necessary, tuned to match experimental data. When the actual time required for thermal or airflow stabilization is of interest, then additional transient CFD simulations should be performed. Such simulations would enable to separate central processing unit computational time from physical central processing unit computational time.

The future development should include the transport of humidity and carbon dioxide, which are directly affecting microclimate of the leafy vegetables. The current enclosure level model also does not fully capture the aerodynamic resistance created by the crop canopy, and this should also be captured in future CFD models. Future models could also incorporate evapotranspiration and plant-specific heat and mass transfer to transition from ventilation assessment to environmental optimization of crops.

Fan energy consumption, operating cost, humidification, carbon dioxide distribution and crop-growth indicators can be added as decision criteria to the optimization framework. When a significantly larger design space is explored,

a multi-objective optimization method may then be combined with the CFD and machine-learning surrogate models, thus decreasing the number of computations required. Systematic variation in criterion weights and the grey relational analysis distinguishing coefficient should also be used to test the robustness of the AHP–grey relational analysis decision framework. This analysis would show whether or not Configuration 1 is still the best alternative when various decision priorities are used.

Therefore, the present study provides a basis for the design and multi-criteria configuration selection of the ventilation in vertical hydroponic systems using CFD. The main achievement lies not only in identifying a specific fan configuration but also in providing a structured approach that allows for evaluating airflow, pressure, thermal behavior and computational aspects simultaneously. Based on the investigated design space and adopted design criteria, the optimal configuration was determined to be the one where one inlet fan and one exhaust fan are used while the sensitivity results indicate that the inlet/exhaust fan capacity ratio is more relevant than the number of fans. Experimental validation, transient modeling, representation of plant canopy, inclusion of other environmental variables and energy-aware optimization are the main directions needed to further validate the applicability and generalizability of the proposed framework.

### Data Availability

Data are available from the author upon reasonable request.

### Conflicts of Interest

The author declares no conflicts of interest.

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